

Mikołaj Karkowski

Full-stack .NET Engineer

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PROFILE

Full-stack .NET engineer with 7+ years of commercial experience building web and desktop products for complex, data-intensive domains. I turn specialized requirements into maintainable software—from real-time Blazor interfaces and ASP.NET Core services to WPF clients and applied AI systems. I value clear boundaries, strong typing, practical automation and close collaboration with users.

CORE SKILLS

- Languages & runtime
C#, .NET 10, T-SQL
- Web
ASP.NET Core, Blazor
WebAssembly, Angular, REST APIs, SignalR
- Desktop
WPF, MVVM, .NET MAUI Hybrid
- Data
EF Core, PostgreSQL, SQL Server, SQLite
- AI
Microsoft.Extensions.AI, OpenRouter, GitHub Models, Ollama, YOLO
- Engineering
DDD, CQRS, modular monoliths, Docker, .NET Aspire, OpenTelemetry, GitHub Actions
- Testing
xUnit, NUnit, Moq, WebApplicationFactory

EDUCATION

- Master of Computer Science
2021–2022
Silesian University of Technology
- Bachelor of Automation and Robotics
2016–2021
Silesian University of Technology

LANGUAGES

- Polish — native
English — C1

INTERESTS

Photography, hiking, travel

EXPERIENCE

- Software DeveloperApr 2019 — present
Graw / Goldschmidt Group
Design, develop and maintain web and desktop .NET products for railway measurement, inspection and navigation systems. Collaborate directly with clients to shape requirements and deliver effective solutions.
 - Built Blazor WebAssembly and .NET MAUI Hybrid interfaces presenting live camera feeds, laser measurements and equipment status with role-based UI access.
 - Created WPF tools for visualizing measurements, reviewing rail defects, displaying equipment diagnostics and working with video-backed image data.
 - Developed an AI-assisted train-navigation algorithm combining video, computer vision, rail infrastructure and operational data as an alternative where GPS is unreliable.
 - Built a rail-object detection tool and trained YOLO models for finding damage on tracks and sleepers.
- Engineering InternApr 2018 — Jun 2018
Bobson
Programmed Arduino boards, tested CAN networks and operated milling equipment.
- Junior Software Developer InternApr 2015
QNT Systems
Developed a responsive web application using Bootstrap.

SELECTED PROJECTS

- FinanceManagerGitHub
Open-source personal finance platform
Full-stack .NET 10 product for managing cash, stocks and bonds with analytics, live market data, CSV workflows and AI-generated insights. Built as a layered modular monolith with a pure domain layer, typed HTTP clients, background channels, SignalR, JWT authentication and OpenTelemetry.
- KotletGitHub
Connected home-kitchen workspace
Recipe, pantry, meal-planning and shopping platform with a shared ingredient catalogue. Includes 28-day planning, nutrition and cost calculations, public recipe sharing, image workflows and AI-assisted imports from video sources.